Explanation of test script:

1) avg_shot_length (Average Shot Length)

What is a "shot"?
A shot is a continuous segment of video from one camera take until the next cut (scene change).

What this feature measures:
The **average duration (in seconds)** of a shot in the video.

How it's computed:

1. The API finds shot boundaries (start/end times).
2. For each shot: length = end - start
3. Average: avg_shot_length = sum(lengths) / number_of_shots

What it tells you:

- **High avg shot length** → fewer cuts, calmer pacing, more stable camera.
- **Low avg shot length** → many cuts, faster pacing, more dynamic editing.

Example intuition:

- A quiet conversation scene → longer shots → higher average
- An action montage → quick cuts → lower average

---

2) shot_variance (Variance of Shot Lengths) (שונות)

Variance describes **how much the shot lengths differ from each other**.

What this feature measures:
Are all shots about the same length, or are some very short and some very long?

How it's computed:

- First compute avg_shot_length
- Then:
  shot_variance = average of (shot_length - avg_shot_length)^2

What it tells you:

- **Low variance** → shots are consistently similar in length
  (steady editing style)
- **High variance** → mixture of short + long shots
  (editing is changing pace)

- **Example intuition:**
- News interview → consistent shot lengths → low variance
- Dramatic scene with quick reaction cuts + long establishing shot → higher variance

---

3) object_frequency_distribution

What it is:
A count of **how often each object type was detected** in the video.

Example output:

```
{
 "person": 12,
 "table": 4,
 "lamp": 2
}
```

Important detail:
In your current implementation, this is counting **object tracks** (each tracked instance over a time interval), not "frames".
So if a person appears in two separate intervals, it may count as 2 tracks.

What it tells you:

- What the video is "about" visually (dominant objects).
- How "busy" the scene is (many different objects detected vs few).

Why it's useful:
This is your raw material for:

- entropy (diversity)
- environment inference (indoor/outdoor, etc.)
- scene-type classification later

---

4) human_presence_ratio

What it is:
The fraction of time that a **person is visible** in the video.

Range:
0.0 to 1.0

- 0.0 = no humans detected at all
- 1.0 = humans visible during the entire video

How it's computed:

1. Collect all time intervals where object = "person" was tracked.
2. Merge overlapping intervals (so you don't double-count time).
3. Add up the total "person visible time".
4. Divide by video duration:

human_presence_ratio = (person_visible_seconds) / (total_video_seconds)

What it tells you:

- High ratio → human-centric scene (dialogue, interview, acting)
- Low ratio → environment-focused (landscape, objects, animals, etc.)

---

5) object_entropy

Entropy is a measure of **diversity / randomness**.

What it is (simple):

- If the same object appears again and again → entropy is **low**
- If many different objects appear with similar frequency → entropy is **high**

How it's computed:
Using Shannon entropy on the object frequency distribution: (Shannon entropy - quantifies the average uncertainty, randomness, or information content in a dataset or stochastic source)

1. Convert counts into probabilities:
    - p(object) = count(object) / total_counts

2. Compute:

H = - Σ p_i * log2(p_i)

What it tells you:

- Low entropy → visually "simple" scene (mostly one object type)
- High entropy → visually "complex" scene (many object types)

Example:

- Video where only "person" appears a lot → low entropy
- Video with person, car, dog, street signs, bicycle, tree… → high entropy

---

6) interaction_density

This one is your "how much is happening" score (based on objects).

Your current definition:
Number of object tracks per second.

interaction_density = total_object_tracks / video_duration_seconds

What it tells you:

- High density → lots of detected object activity/appearance
- Low density → fewer tracked objects, simpler scene

Important note:
This doesn't *guarantee* "interaction between people" — it's more like **visual activity density**.
Later, if you want a more "interaction-specific" metric, you can evolve it into:

- "number of distinct objects overlapping in time with person"
- "multi-object co-occurrence rate"
- "person + object overlap ratio"

Summary in one sentence each (quick memory)

- **avg_shot_length**: how long shots are on average → pacing
- **shot_variance**: how consistent shot lengths are → pacing stability
- **object_frequency_distribution**: which objects appear most → visual content
- **human_presence_ratio**: how much time humans are visible → human-centric vs environment
- **object_entropy**: diversity of detected objects → scene complexity
- **interaction_density**: objects tracked per second → how "busy" the video is